KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 22 · MATH

Ship Day in Duluth Harbor

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Measuring the Harbor

Solveig is spending the day with her Uncle Torvald on a tugboat in Duluth, Minnesota. Warm up before the lesson and show your work. Some problems may feel new, and that is okay!

1. Review (Week 21): a) Write the ordered pair for a point 4 units right of the origin and 6 units up. b) Rule A: start at 0, add 2. Rule B: start at 0, add 10. Which Rule B number corresponds to 6 in Rule A?

2. Fill in the blanks: 1 foot = _____ inches. 1 yard = _____ feet. 1 yard = _____ inches. 1 pound = _____ ounces.

3. Solveig has a rope that is 3 yards long. How many feet long is it? How many inches long is it?

4. Which unit makes the most sense for the mass of a crate of bolts?

   kilograms | milliliters | kilometers

5. 1 gallon = 4 quarts. How many quarts are in 2 1/2 gallons? Draw a picture to show your thinking.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: The Harbor Crew

Bigger unit to smaller unit, multiply. Smaller unit to bigger unit, divide. Show your work and label every answer.

1. a) 3 2/3 yards = _____ feet b) 1 3/4 feet = _____ inches c) 95 inches = _____ feet _____ inches

2. a) 2 3/4 pounds = _____ ounces b) 6 gallons = _____ cups c) 9 pints = _____ quarts (use a mixed number)

3. a) 4.8 kilometers = _____ meters b) 560 centimeters = _____ meters c) 3,250 milliliters = _____ liters

4. Which is heavier: a 1 3/4-pound bag of washers or a 30-ounce bag of washers?

   1 3/4 pounds | 30 ounces | They weigh the same

5. Find the mistake: Sylvie says 2.6 liters = 26 milliliters. In two or three complete sentences, explain her mistake and give the correct answer.

★ **Challenge:** Uncle Torvald has a 40-yard tow line. He cuts 6 pieces that are each 14 feet long. How many feet of line are left? Write the answer in yards, too.

Looking Ahead: Week 23: The Tide Pool Survey. Make and read line plots with fractional data in halves, fourths, and eighths.